"Ridgeline HD: Front clunk and uneven tire wear from worn lower control arm

Document: tsb-20-508
Area: bulletins

1 Condition

Ridgeline HD operators may report a metallic clunk from the front suspension when crossing bumps or during direction changes, frequently accompanied by uneven inner-edge tire wear. The noise and wear both trace to play in the front lower control arm (CARM-8540) bushings and ball joint. Verify the complaint on a hoist by checking for free play at the control arm before proceeding. See the Ridgeline HD Suspension Service Manual :: Control Arms and Sway Bar Links for inspection criteria.

2 Cause

The clunk and accelerated tire wear are caused by deterioration of the lower control arm (CARM-8540) bushings, which allows the arm to deflect under load and shifts the front wheel alignment toward excessive negative camber and toe-out. Once the bushing or integrated ball joint exceeds the play specification, the arm cannot be serviced separately and must be replaced as an assembly. Review the Ridgeline HD Suspension Service Manual :: Suspension System Overview to confirm the affected components before ordering parts.

3 Repair Procedure

Replace the worn lower control arm (CARM-8540) following PROC-CONTROL-ARM, observing the specified fastener torque and the requirement to torque bushing bolts at curb height. After installation, perform a four-wheel alignment per PROC-ALIGN to restore camber and toe and to prevent recurrence of the inner-edge tire wear. The detailed removal and installation steps are in the Ridgeline HD Suspension Service Manual :: Control Arm Replacement Procedure , and the alignment is documented in the Lumen Steering Service Manual :: Four-Wheel Alignment Procedure . For sub-system context, see the Ridgeline HD Suspension Service Manual :: Control Arm and Sway Bar Sub-System Service .